

Django Models - Actual Database Roadmap

System Overview: Total 11 Internal Models

This roadmap is strictly generated by analyzing your codebase. Every single database field has been given its own dedicated row below for clarity.

Table Relationships & Connections:

- **Driver Table:** Primary Key: driver_id. No foreign keys. Referenced by Vehicle.driver (FK), Trip.driver (FK), LRBilty.driver (FK), EWayBill.driver (FK).
- **Vehicle Table:** Primary Key: vehicle_id. Foreign Key: driver → Driver.driver_id. Referenced by Fuel.vehicle (FK), Toll.vehicle (FK), Trip.vehicle (FK), LRBilty.vehicle (FK), EWayBill.vehicle (FK), FinanceTransaction.vehicle (FK), Tracking.vehicle (FK).
- **Fuel Table:** Primary Key: fuel_id. Foreign Key: vehicle → Vehicle.vehicle_id. No references.
- **Toll Table:** Primary Key: toll_id. Foreign Key: vehicle → Vehicle.vehicle_id. No references.
- **Trip Table:** Primary Key: trip_id. Foreign Keys: vehicle → Vehicle.vehicle_id, driver → Driver.driver_id. Referenced by FinanceTransaction.trip (FK).
- **LRBilty Table:** Primary Key: lr_id. Foreign Keys: vehicle → Vehicle.vehicle_id, driver → Driver.driver_id, eway_bill → EWayBill.eway_id. Referenced by EWayBill.lr (FK).
- **EWayBill Table:** Primary Key: eway_id. Foreign Keys: lr → LRBilty.lr_id, vehicle → Vehicle.vehicle_id, driver → Driver.driver_id. No references.
- **FinanceTransaction Table:** Primary Key: transaction_id. Foreign Keys: vehicle → Vehicle.vehicle_id, trip → Trip.trip_id, vendor → Vendor.vendor_id. No references.
- **Vendor Table:** Primary Key: vendor_id. No foreign keys. Referenced by FinanceTransaction.vendor (FK), Inventory.vendor (FK).
- **Inventory Table:** Primary Key: item_id. Foreign Key: vendor → Vendor.vendor_id. No references.
- **Tracking Table:** Primary Key: tracking_id. Foreign Key: vehicle → Vehicle.vehicle_id. No references.
- **SystemSetting Table:** Primary Key: setting_id. No foreign keys. No references.

Null/Not Null Summary: Primary keys are NOT NULL. Required fields (without blank=True/null=True) are NOT NULL. Optional fields are NULL allowed. Foreign keys can be NULL if not assigned.

1. DRIVER Table

Column (Field) Name	Data Type	Constraints & Default Rules	Key/Connection
driver_id	AutoField	NOT NULL (Primary Key)	PK - Unique identifier
name	CharField	NULL/Blank Allowed	-

license	CharField	NULL/Blank Allowed	-
phone	CharField	NULL/Blank Allowed	-
altPhone	CharField	NULL/Blank Allowed	-
experience	CharField	NULL/Blank Allowed	-
address	TextField	NULL/Blank Allowed	-
state	CharField	NULL/Blank Allowed	-
city	CharField	NULL/Blank Allowed	-
aadhar	CharField	NULL/Blank Allowed	-
photo	ImageField	NULL/Blank Allowed	-
dob	DateField	NULL/Blank Allowed	-
age	PositiveIntegerField	NULL/Blank Allowed	-
medical	TextField	NULL/Blank Allowed	-
maritalStatus	CharField (Choices)	NULL/Blank Allowed	-
nationality	CharField	NULL/Blank Allowed	-
jobType	CharField (Choices)	NULL/Blank Allowed	-
status	CharField (Choices)	NOT NULL. Default: "Active"	-
added_date	DateTimeField	Auto-populated. NULL/Blank Allowed	-
updated_date	DateTimeField	Auto-populated. NULL/Blank Allowed	-

2. FUEL Table

Column (Field) Name	Data Type	Constraints & Default Rules	Key/Connection
fuel_id	AutoField	NOT NULL (Primary Key)	PK - Unique identifier
vehicle	ForeignKey (Vehicle)	NULL Allowed	FK - Connects to Vehicle.vehicle_id

litres	FloatField	NOT NULL	-
price_per_litre	FloatField	NOT NULL	-
date	DateField	NOT NULL	-
photo	ImageField	NULL/Blank Allowed	-
added_date	DateTimeField	Auto Now Add	-

3. TOLL Table

Column (Field) Name	Data Type	Constraints & Default Rules	Key/Connection
toll_id	AutoField	NOT NULL (Primary Key)	PK - Unique identifier
vehicle	ForeignKey (Vehicle)	NULL Allowed	FK - Connects to Vehicle.vehicle_id
toll_name	CharField	NOT NULL	-
amount	FloatField	NOT NULL	-
date	DateField	NOT NULL	-
photo	ImageField	NULL/Blank Allowed	-
added_date	DateTimeField	Auto Now Add	-

4. VEHICLE Table

Column (Field) Name	Data Type	Constraints & Default Rules	Key/Connection
vehicle_id	AutoField	NOT NULL (Primary Key)	PK - Unique identifier
vehicle_number	CharField	NULL/Blank Allowed	-
make	CharField	NULL/Blank Allowed	-
model	CharField	NULL/Blank Allowed	-
capacity	CharField	NULL/Blank Allowed	-
driver	ForeignKey (Driver)	NULL Allowed	FK - Connects to Driver.driver_id

status	CharField (Choices)	NOT NULL. Default: 'Available'	-
added_date	DateTimeField	Auto Now Add	-

5. TRIP Table

Column (Field) Name	Data Type	Constraints & Default Rules	Key/Connection
trip_id	AutoField	NOT NULL (Primary Key)	PK - Unique identifier
vehicle	ForeignKey (Vehicle)	NULL Allowed	FK - Connects to Vehicle.vehicle_id
driver	ForeignKey (Driver)	NULL Allowed	FK - Connects to Driver.driver_id
start_location	CharField	NULL/Blank Allowed	-
end_location	CharField	NULL/Blank Allowed	-
start_time	DateTimeField	NULL/Blank Allowed	-
end_time	DateTimeField	NULL/Blank Allowed	-
status	CharField (Choices)	NOT NULL. Default: 'Scheduled'	-
distance	FloatField	NOT NULL. Default: 0	-
fuel_consumed	FloatField	NOT NULL. Default: 0	-
added_date	DateTimeField	Auto Now Add	-

6. LRBILTY Table

Column (Field) Name	Data Type	Constraints & Default Rules	Key/Connection
lr_id	AutoField	NOT NULL (Primary Key)	PK - Unique identifier
lr_number	CharField	NULL/Blank Allowed	-
date	DateField	NULL/Blank Allowed	-

consignor	CharField	NULL/Blank Allowed	-
consignee	CharField	NULL/Blank Allowed	-
route	CharField	NULL/Blank Allowed	-
vehicle	ForeignKey (Vehicle)	NULL Allowed	FK - Connects to Vehicle.vehicle_id
driver	ForeignKey (Driver)	NULL Allowed	FK - Connects to Driver.driver_id
material	CharField	NULL/Blank Allowed	-
weight	CharField	NULL/Blank Allowed	-
freight	CharField	NULL/Blank Allowed	-
eway_bill	ForeignKey (EWayBill)	NULL Allowed	FK - Connects to EWayBill.eway_id
status	CharField (Choices)	NOT NULL. Default: 'Pending'	-
added_date	DateTimeField	Auto Now Add	-

7. EWAYBILL Table

Column (Field) Name	Data Type	Constraints & Default Rules	Key/Connection
eway_id	AutoField	NOT NULL (Primary Key)	PK - Unique identifier
invoice_number	CharField	NULL/Blank Allowed	-
lr	ForeignKey (LRBilty)	NULL Allowed	FK - Connects to LRBilty.lr_id
supplier_name	CharField	NULL/Blank Allowed	-
supplier_gstin	CharField	NULL/Blank Allowed	-
buyer_name	CharField	NULL/Blank Allowed	-
buyer_gstin	CharField	NULL/Blank Allowed	-
goods_description	CharField	NULL/Blank Allowed	-
hsn_code	CharField	NULL/Blank Allowed	-

invoice_amount	CharField	NULL/Blank Allowed	-
vehicle	ForeignKey (Vehicle)	NULL Allowed	FK - Connects to Vehicle.vehicle_id
route_from	CharField	NULL/Blank Allowed	-
route_to	CharField	NULL/Blank Allowed	-
estimated_days	IntegerField	NULL/Blank Allowed. Default: 0	-
driver	ForeignKey (Driver)	NULL Allowed	FK - Connects to Driver.driver_id
driver_phone	CharField	NULL/Blank Allowed	-
status	CharField (Choices)	NOT NULL. Default: 'Active'	-
added_date	DateTimeField	Auto Now Add	-

8. FINANCETRANSACTION Table

Column (Field) Name	Data Type	Constraints & Default Rules	Key/Connection
transaction_id	AutoField	NOT NULL (Primary Key)	PK - Unique identifier
date	DateField	NULL/Blank Allowed	-
type	CharField (Choices)	NOT NULL. Default: 'Income'	-
description	TextField	NULL/Blank Allowed	-
amount	DecimalField	NOT NULL. Default: 0.0	-
status	CharField (Choices)	NOT NULL. Default: 'Completed'	-
category	CharField	NULL/Blank Allowed	-
vehicle	ForeignKey (Vehicle)	NULL Allowed	FK - Connects to Vehicle.vehicle_id
trip	ForeignKey (Trip)	NULL Allowed	FK - Connects to Trip.trip_id

vendor	ForeignKey (Vendor)	NULL Allowed	FK - Connects to Vendor.vendor_id
added_date	DateTimeField	Auto Now Add	-

9. VENDOR Table

Column (Field) Name	Data Type	Constraints & Default Rules	Key/Connection
vendor_id	AutoField	NOT NULL (Primary Key)	PK - Unique identifier
name	CharField	NOT NULL	-
service_type	CharField	NULL/Blank Allowed	-
contact_person	CharField	NULL/Blank Allowed	-
phone	CharField	NULL/Blank Allowed	-
email	EmailField	NULL/Blank Allowed	-
address	TextField	NULL/Blank Allowed	-
status	CharField (Choices)	NOT NULL. Default: 'Active'	-
added_date	DateTimeField	Auto Now Add	-

contact_personCharFieldNULL/Blank Allowed phoneCharFieldNULL/Blank Allowed emailEmailFieldNULL/Blank Allowed addressTextFieldNULL/Blank Allowed statusCharField (Choices)NOT NULL. Default: 'Active' added_dateDateTimeFieldAuto Now Add

10. INVENTORY Table

Column (Field) Name	Data Type	Constraints & Default Rules	Key/Connection
item_id	AutoField	NOT NULL (Primary Key)	PK - Unique identifier
name	CharField	NOT NULL	-
category	CharField	NULL/Blank Allowed	-
quantity	IntegerField	NOT NULL. Default: 0	-
unit_price	DecimalField	NOT NULL. Default: 0.0	-

vendor	ForeignKey (Vendor)	NULL Allowed	FK - Connects to Vendor.vendor_id
status	CharField (Choices)	NOT NULL. Default: 'In Stock'	-
added_date	DateTimeField	Auto Now Add	-

11. TRACKING Table

Column (Field) Name	Data Type	Constraints & Default Rules	Key/Connection
tracking_id	AutoField	NOT NULL (Primary Key)	PK - Unique identifier
vehicle	ForeignKey (Vehicle)	NULL Allowed	FK - Connects to Vehicle.vehicle_id
current_location	CharField	NULL/Blank Allowed	-
latitude	DecimalField	NULL/Blank Allowed	-
longitude	DecimalField	NULL/Blank Allowed	-
speed	FloatField	NOT NULL. Default: 0.0	-
status	CharField (Choices)	NOT NULL. Default: 'Moving'	-
timestamp	DateTimeField	Auto Now Add	-

12. SYSTEMSETTING Table

Column (Field) Name	Data Type	Constraints & Default Rules	Key/Connection
setting_id	AutoField	NOT NULL (Primary Key)	PK - Unique identifier
key	CharField	NOT NULL. Unique	-
value	TextField	NOT NULL	-
description	TextField	NULL/Blank Allowed	-
updated_date	DateTimeField	Auto Now	-

setting_idAutoFieldNOT NULL (Primary Key) keyCharFieldNOT NULL. UNIQUE valueTextFieldNOT NULL
descriptionTextFieldNULL/Blank Allowed updated_dateDateTimeFieldAuto Now